Telco Customer Churn Prediction using Random Forest

Random Forest Machine Learning Project

Dataset

Telco Customer Churn (Kaggle, by BlastChar / IBM) – kaggle.com/datasets/blastchar/telco-customer-churn. Real customer data: 7,043 rows, 21 columns (demographics, services, contract, billing, churn label).

Problem Statement

Predict whether a telecom customer will churn (leave the company) so the business can target retention offers. This is a binary classification task. The data is imbalanced: 26.5% churn vs 73.5% stay.

Data Preprocessing Steps

- Found 11 hidden blank values in *TotalCharges* (stored as text); converted to numeric and filled with 0 (these customers have tenure = 0, i.e. not yet billed).
- Dropped *customerID* (identifier, no predictive value).
- Encoded target (Yes=1, No=0) and one-hot encoded all categorical columns (*drop_first=True*), giving 30 features.
- Split features/target; 80/20 stratified train-test split (random_state=42): 5,634 train / 1,409 test rows.

Model Used

RandomForestClassifier (scikit-learn). Baseline: 200 trees. Tuned with GridSearchCV (3-fold, optimising F1): *n_estimators=400, max_depth=8, min_samples_leaf=5, class_weight='balanced'*. 5-fold cross-validation and model saving (joblib) included as bonus.

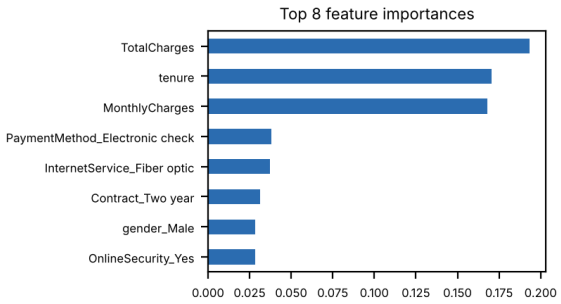
Evaluation Metrics (test set)

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Baseline RF	0.793	0.637	0.511	0.567	0.826
Tuned RF	0.752	0.522	0.797	0.631	0.844

5-fold CV accuracy (baseline): 0.786 ± 0.007. A confusion matrix is included in the notebook.

Key Findings

- Top features: **TotalCharges, tenure, MonthlyCharges**, then electronic-check payment, fibre-optic internet and two-year contract.
- Short-tenure customers with high monthly bills, fibre-optic service and electronic-check payment are most likely to churn.
- Long-term (1–2 year) contracts are linked to lower churn.
- The baseline misses ~49% of churners (recall 0.51). Balancing class weights raises recall to 0.80 and F1 to 0.63 at the cost of some accuracy.



Conclusion

Random Forest works well for churn prediction on this dataset (ROC-AUC 0.84). Because churners are a minority, accuracy alone is misleading; the tuned, class-balanced model is preferred since it catches ~80% of churners, which suits a retention campaign. The business should push new customers toward longer contracts and target high-bill, short-tenure customers with offers. Future work: gradient boosting, SMOTE and decision-threshold tuning.